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The Government as Venture Capitalist: The Long-Run Impact of the SBIR Program*

I. Introduction

Despite economists' interest in interactions between governments and firms, the public subsidization of small firms has attracted virtually no scrutiny. This article seeks to address this omission by examining the largest such program in the United States, the Small Business Innovation Research (SBIR) program, which has provided over \$7 billion to small high-technology firms between 1983 and 1997.

Addressing this omission is important for three reasons:

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Government programs to finance small firms have attracted little empirical attention. This article examines the largest U.S. initiative, the Small Business Innovation Research program. Using a unique database, I show that program awardees grew significantly faster than matched firms over a decade and were more likely to attract venture financing. The superior performance of awardees was confined to firms in regions with substantial venture capital activity and was pronounced in high-technology industries. Multiple awards did not increase performance. These results suggest that awards played an important role in certifying firm quality but also that distortions of the award process occur.

First, it provides an opportunity to examine empirically theoretical models of interactions between the public and private sectors. This has been a topic of enduring interest in the economics literature for several decades.

Second, public programs to subsidize small high-technology firms have represented a significant but little-studied area of public expenditures. (Table 1 lists the programs in the United States over the past several decades. See Organization for Economic Cooperation and Development [1996] for tabulations of efforts in Europe and Asia.) For instance, the sum of the financing provided through, and guaranteed by, the programs listed in table 1 in 1995 was at least \$2.4 billion, compared to the \$3.9 billion disbursed by private venture capital funds in that year.

Finally, a number of these programs are reputed to have had a positive and significant effect on economic growth. Some of America's most dynamic companies received support through federal programs while privately held entities, including Apple Computer, Chiron, Compaq, Federal Express, and Intel. In addition to funding firms, publicly sponsored funds during the 1960s provided early experience for many of the individuals who later went on to lead independent venture organizations. Overseas, much of the recent growth in high-technology firms in such nations as Israel, Singapore, and Taiwan has been attributed to government venture capital initiatives (see Organization for Economic Cooperation and Development 1996).

Economic analyses suggest at least two sets of rationales for governments to offer subsidies to small high-technology firms. First, the social returns from the firms' R & D expenditures may exceed their private returns. Second, in making such awards, knowledgeable government officials may certify firms to private investors and address the informational asymmetries that might have otherwise precluded investments. An extensive literature on regulatory capture, however, suggests that government involvement may be distorted by the desire of interest groups—or of the politicians themselves—to maximize their own private benefits.

This article assesses the long-run success of firms participating in the SBIR program. I examine the employment and sales growth of 1435 firms over a 10-year period. Two-fifths of the sample received one or more awards in the first three cycles of the program; the others were matching firms chosen to closely resemble the awardees. The analysis relied heavily on a unique data set of awardees compiled by the U.S. General Accounting Office (GAO).

Over this period, the SBIR awardees enjoyed substantially greater employment and sales growth than the matching firms. The growth of the awardees did not appear to have been due to the receipt of procurement contracts by these firms or other alternative explanations. These

TABLE 1 U.S. Public Venture Capital Initiatives, 1958-97

Sponsoring Organization	Program Name	Brief Description	Span
Small Business Administration	Small Business Investment Company Program	Provides capital to federally sponsored funds that make debt and equity investments in growth firms.	1958-97
Department of Commerce	State Technical Services Program	Supported various government programs to help high-technology companies (especially new firms).	1965-69
Department of Housing and Urban Development, Model Cities Administration	Venture Capital Development Assistance	Demonstration projects in selected cities financed businesses begun by residents of targeted neighborhoods.	1967-71
At least 30 states	At least 43 state venture funds or SBIC programs	Make investments into funds supporting new enterprises, which often focus on high-technology firms.	1970-97
Department of State, Agency for International Development	At least 13 developing country venture funds	Provided loans to financial intermediaries that made equity and debt investments in new enterprises in over 30 countries.	1971-93
Small Business Administration	Specialized Small Business Investment Company Program	Provides capital to federally sponsored funds that make debt and equity investments in growth firms owned by disadvantaged individuals.	1972-97
Department of Commerce, National Bureau of Standards	Experimental Technology Incentives Program	Catalyzed new public programs (across agencies) to encourage industrial research and venture capital.	1972-79
National Science Foundation	Federal Laboratories Validation Assistance Experiment	Funded assessments by national laboratory personnel of prototype products and processes developed by entrepreneurs.	1972-75
National Science Foundation and Small Business Administration	Innovation Centers Experiment	Provided assistance to high-tech entrepreneurs through incubation centers, subsidies, and technical assistance.	1973-81
Department of Energy, Office of Energy-Related Inventions	Energy Related Inventions Program	Provides financing to individual inventors and small firms to commercialize energy-conserving discoveries.	1975-97

TABLE 1 (Continued)

Sponsoring Organization	Program Name	Brief Description	Span
Small Business Administration	Small Business Development Centers Program	Funds university-based centers to assist small businesses and encourage technology transfer.	1976-97
Department of Commerce	Corporations for Innovation Development Initiative	Designed to fund state and regional corporations to provide equity financing to new firms; only one such corporation funded.	1979-81
Department of Commerce, Minority Business Development Agency	Technology Commercialization Program	Financed minority technology-oriented entrepreneurs, as well as centers to assist such entrepreneurs.	1979-82
At least 15 states	At least 107 business incubators	Provide office and manufacturing space, support services, and often financing to start-up businesses.	1980-96
11 federal agencies	Small Business Innovation Research Program	Provides awards to small technology-oriented businesses (also predecessor programs at 3 agencies, 1977-82).	1982-97
Department of Energy, Office of Energy Research	At least 6 contractor-organized venture funds	Make equity investments in spinouts from national laboratories (funds organized by prime or subcontractors at laboratories with Department's encouragement).	1985-97
At least 30 states	State Small Business Innovation Research Programs	Make SBIR-like grants, often in conjunction with federal SBIR awards.	1987-97
Department of Commerce, National Institute of Standards and Technology	Advanced Technology Program	Awards grants to develop targeted technologies to firms and consortia; some emphasis on small businesses.	1988-97
Department of Defense, Defense Advanced Research Projects Agency	Experimental venture capital investment program	Designed to make investments in private high-technology firms in exchange for equity or royalties; program only made one investment.	1989-91

Department of State, Agency for International Development	Enterprise Fund Program	Oversees 12 federally funded venture funds investing in Eastern Europe, the former Soviet Union, and Africa.	1990-97
Overseas Private Investment Corporation	Venture capital fund guarantees	Guarantees full or partial return of capital to investors in at least 16 private venture funds in developing countries.	1990-97
Department of Housing and Urban Development, Community Relations and Involvement Office	Tenant Opportunity Program	Funds new businesses and other initiatives by public housing residents (other aspects of program had begun in 1987).	1993-97
Department of Energy, Office of the Under Secretary	Defense Programs Small Business Initiative	Provides funding, technological assistance, and national laboratory access to small high-technology businesses.	1993-97
11 federal agencies	Small Business Technology Transfer Program	Finances cooperative research projects between small high-technology firms and non-profit research institutions.	1994-97
Department of Defense, Cooperative Threat Reduction Program	Defense Enterprise Fund	Finances an independent venture fund investing in defense conversion projects in the former Soviet Union.	1994-97
Department of the Treasury	Community Development Financial Institutions Fund	Invests in and provides assistance to community development venture capital and loan funds.	1995-97
Department of Defense	"Fast Track" Program	Provides matching funds for private financing raised by SBIR awardees.	1995-97
Department of Agriculture, Rural Business and Cooperative Development Service	Intermediary Relending Program (as amended)	Permits program managers to guarantee returns of investors in rural venture funds.	1997

SOURCE.—Eisinger (1988); Organization for Economic Cooperation and Development (1996); U.S. Small Business Administration (1996); and various news stories in the LEXIS/NEXIS database and government reports.

NOTE.—The table summarizes programs sponsored by state and federal organizations in which equity investments or equity-like grants were made into privately held companies or into funds that made such investments. If a program had multiple names, I report the name in May 1997. If a program was terminated prior to this date, I record its name at the time of the termination. If an organization sponsoring a program changed its name, or if responsibility for the program was transferred between organizations, I report the name of the sponsoring organization in May 1997. If the program was terminated prior to this date, I record the sponsoring organization at the time of the program's termination.

firms were also more likely to subsequently receive venture capital financing. The growth, however, was not uniform. The superior performance was confined to awardees in areas with substantial new firm creation, as measured by early-stage venture capital disbursements. The relationship between SBIR awards and growth was much stronger in high-technology industries. The SBIR awardees receiving large subsidies did not perform better than those receiving smaller subsidies. The results suggest that these awards played an important certification function. At the same time, distortions in the award process may have adversely affected the program's effectiveness, especially in regions with fewer high-technology firms and among firms receiving large subsidies.

The analysis had two important limitations. The most critical of these was that I was not able to assess the effect of these programs on social welfare. Only with much more detailed data could the relative costs and benefits of the initiative be fully assessed. Second, my focus on a single program limited my ability to discuss the effect of alternative program designs.

The plan of the article is as follows. In Section II, I discuss economic perspectives on the subsidization of small high-technology firms and their empirical implications. The third section reviews the structure of the SBIR program and the key insights gained through interviews. Section IV discusses the challenges associated with empirically evaluating such programs. The fifth section describes the data set. Section VI presents the analysis, and the final section concludes the article.

II. The Subsidization of Small High-Technology Firms

A. Theoretical Motivations

The economics literature suggests at least two sets of rationales for governments to offer subsidies to small high-technology firms. Other works, however, argue that government involvement may be distorted by the desire of interest groups or politicians to maximize their own benefits. These suggest a more skeptical view of such programs.

Public finance theory emphasizes that subsidies are an appropriate response to activities that generate positive externalities. Such investments as R & D expenditures and pollution control equipment purchases may have positive spillovers that benefit other firms or society as a whole. But because the firms making these investments are unlikely to capture the entire surplus, public subsidies may be an appropriate response.

An extensive literature (reviewed in Griliches [1992] and Jaffe [1996]) has documented the presence of R & D spillovers. These spillovers take several forms. The profits associated with new innovations

may accrue to competitors who rapidly introduce imitations, to developers of complementary products, or to consumers of the new products. Whatever the mechanism of the spillovers, the consequence is the same: firms will tend to invest below the socially optimal level of R & D. After reviewing a wide variety of studies, Griliches argues that the social rate of return exceeds the private rate by a considerable amount. He estimates that the social rate of return is between 150% and 200% of the private rate of return. Few studies have examined how these gaps vary with firm characteristics. A number of case-based analyses (Jewkes, Sawers, and Stillerman 1958; Mansfield et al. 1977), however, suggest that spillover problems are particularly severe among small firms, which are often unable to defend effectively their intellectual property or to extract most of the rents in the product market.

A second rationale for public subsidies for small high-technology firms lies in the fact that they may convey information to other potential investors. Informational asymmetries may make raising external capital very expensive for entrepreneurs or even preclude it entirely. Myers and Majluf (1984) and Greenwald, Stiglitz, and Weiss (1984) demonstrate that equity offerings of firms may be associated with a "lemons" problem. If the manager is better informed about the investment opportunities of the firm and acts in the interest of current shareholders, then he only issues new shares when the company's stock is overvalued. Indeed, numerous studies have documented that stock prices decline on the announcement of equity issues, largely because of the negative signal that it sends to the market. Information problems have also been shown to exist in debt markets by Stiglitz and Weiss (1981).

These problems are a consequence of the information gaps between the entrepreneurs and investors. If the information asymmetries could be eliminated, financing constraints would disappear. Financial economists argue that specialized financial intermediaries, such as venture capital organizations, can address these problems by intensively scrutinizing firms before providing capital and then monitoring them afterward. Nonetheless, venture capitalists back only a tiny fraction of the technology-oriented businesses begun each year. In 1997, a record year for venture disbursements, 707 companies received venture financing for the first time.¹ By way of comparison, the Small Business Administration estimates that 885 thousand businesses were started in 1997. If federal awards could effectively certify firms to their potential investors, these problems could be addressed.²

The arguments outlined above implicitly assume that, once a market

1. Summary data on this and other years are available at <http://www.ventureone.com>.

2. Consistent with this view, several directories of high-technology firms denote companies that have won SBIR awards, and at least 167 press releases concerning SBIR awards were issued by firms in 1996.

failure is identified, the government can dispassionately address it. An extensive political economy and public finance literature, however, has emphasized the distortions that may result from government subsidies, as interest groups or politicians seek to direct subsidies in a manner that benefits themselves. As articulated by Olson (1965) and Stigler (1971), and formally modeled by Peltzman (1976) and Becker (1983), the theory of regulatory capture suggests that direct and indirect subsidies will be captured by groups that stand to gain substantial benefits and whose collective political activity is not too difficult to arrange (i.e., when "free-riding" by coalition members is not too large a problem). As Stigler (1971) points out, even very small firms (which have historically dominated industries such as trucking and the licensed professions) can organize to benefit from public largess. Many of the initial works in this literature treated politicians and government bureaucrats as passive implementers of policies desired by private coalitions. More recent models (summarized in Laffont and Tirole [1993]) have incorporated voters, politicians, and/or bureaucrats as distinct actors, often in a principal-agent relationship.

These distortions may manifest themselves in several ways. One possibility (discussed, e.g., in Eisinger [1988]) is that firms may seek transfer payments that directly increase their profits. Politicians may acquiesce to such transfers to politically connected companies. A more subtle problem is discussed by Cohen and Noll (1991) and Wallsten (1996): officials may seek to select firms based on their likely success, regardless of whether the government funds are needed. In this case, they can claim credit for the firms' ultimate success, even if the marginal contribution of the public funds was very low.

B. Empirical Implications

These contrasting views have different empirical implications for the relative performance of small high-technology firms receiving public subsidies. In this section, I review four sets of predictions about the effect of public subsidies on firm performance.

Overall effect of subsidies. The first rationale was that subsidies are required to encourage small firms to undertake research because of spillover problems. In this case, subsidy recipients should perform at least slightly better than their peers, even if much of the surplus created by their R & D accrues to other firms in their industry or to consumers. Similarly, if public awards can certify firms to investors, the awardees may be able to attract additional capital and grow more rapidly. If, however, these awards are captured by politically connected firms, the funds may be squandered and not translated into long-run growth. The final view delineated above—that federal agencies are selecting firms that are likely to grow rapidly in any case—also suggests that SBIR firms should perform better than a set of matched firms.

Variation in subsidy effect across regions. The externality and certification views suggest no clear implications for where the regional benefits of the program should be greatest. The more political economy-based views, however, raise the possibility of different geographic effects. In particular, it may be that, in regions without large numbers of high-technology firms and venture capital organizations, these awards may be captured by other firms than the intended population: politicians may encourage awards to politically connected firms with few growth prospects (e.g., consulting and contract research firms). Presumably, these types of distortions in the selection process are much less likely in regions with active communities of high-technology firms and venture investors, as these groups are likely to have a considerable degree of political influence. This suggests that the relative effect of the program may be greater in regions with concentrated populations of venture capitalists and high-technology firms.³

Variations of subsidy effect with industry characteristics. Industries vary tremendously in their degree of transparency to outside investors. In some cases, the financial statements traditionally analyzed by bankers and accountants provide much valuable information. In other instances, the firms' primary assets are intangible intellectual property, which does not show up on balance sheets and may yield returns only after many years. These differences can again be exploited in the empirical analysis:

- The certification hypothesis suggests that if public awards convey information to investors, these signals are likely to be particularly valuable in technology-intensive industries where traditional financial measures are of little use. Specialists at the National Institutes of Health or Department of Defense may have considerable insight into which biotechnology or advanced materials companies are the most promising, while the traditional financial statement analysis undertaken by bankers would be of little value. In these cases, the signal that a government award provides might be particularly significant.
- The hypothesis that federal officials are selecting firms likely to perform in a superior fashion suggests a different pattern. Even though the insights of federal officials may give them an advantage relative to private investors (and thus make an award a valuable signal), they may not find the selection of successful firms in high-technology industries to be easy. Predicting firm success in high-technology in-

3. A natural question is why any awards are made in regions with few high-technology firms. While this issue has not been discussed at great length in the public finance literature, a substantial body of work on political institutions documents that even in programs that disproportionately benefit particular regions (such as Air Force procurement contracts), some federal funds are spread across a wide number of states to insure congressional support. See Carsey, Rundquist, and Fox (1997) for a review of this literature.

dustries is very difficult. For instance, Audretsch (1995, table 4.11) analyzes the growth of 7,000 start-ups over a 10-year period. Both firm and industry characteristics have considerable explanatory power in his regressions analyzing the success of firms in low- and medium-technology industries. In a regression analysis of high-technology firms, however, the coefficients are individually and jointly statistically insignificant. Thus, if the crucial criterion for selecting awardees is the ultimate success of the firm, it is not clear that federal officials will be particularly successful in high-technology industries. I anticipated that federal officials should be equally able—or even more likely—to choose successful firms in low-technology industries. This hypothesis suggests that when comparing SBIR awardees to a sample of matching firms, the superior performance of awardees should not be confined to high-technology industries. In fact, the relative success of low-technology awardees—where identifying firms that will ultimately be successful is substantially easier—may be even greater.

Variations of subsidy effect with the amount of subsidies received. If the main role of public subsidies is to address the externality posed by small firms' inability to capture the bulk of the surplus generated by their innovations, there should be a positive relationship between the amount of subsidy received and the effect on the firm. As long as firms undertaking R & D capture at least some of the benefits from their research, they should grow more quickly as subsidies increase. The certification hypothesis suggests that there may not be a positive relationship between the amount of subsidies and firm growth. In particular, the marginal value of the signal that the tenth subsidy provides should be considerably lower than that of the first. Meanwhile, the pursuit of these awards may become increasingly costly to the firm as the size (or number) of subsidies increases and management attention is diverted from commercialization.

III. The Structure and Implementation of the SBIR Program

The Small Business Innovation Development Act, enacted by Congress in July 1982, established the SBIR program. The program mandated that all federal agencies spending more than \$100 million annually on external research set aside 1.25% of these funds for awards to small businesses. When the program was reauthorized in 1992, Congress increased the size of the set-aside to 2.5%. In 1997, this represented annual funding of about \$1.1 billion.

While the eleven federal agencies participating in the program are responsible for selecting awardees, they must conform to the guidelines stipulated by the act and the U.S. Small Business Administration.

TABLE 2 Volume of SBIR Awards

Year	SBIR Awards		Size of External R & D Set- Aside (%)
	\$ Amount	No. of Phase II Awards	
1983	.07	0	.2
1984	.15	388	.6
1985	.27	407	1.0
1986	.40	564	1.25
1987	.45	768	1.25
1988	.49	711	1.25
1989	.52	749	1.25
1990	.53	837	1.25
1991	.53	788	1.25
1992	.55	916	1.25
1993	.71	950	1.5
1994	.71	961	1.5
1995	.86	1,263	2.0
1996	.89	1,191	2.0
1997	1.08	1,404	2.5

SOURCE.—U.S. Small Business Administration (1996); and personal communications with SBA officials.

NOTE.—The table indicates the amount of SBIR awards (in billions of 1995 dollars), the number of Phase II awards, and the share of external R & D spending that all agencies spending more than \$100 million on external R & D were required to set aside for the program. The subsidy was phased in over a number of years. The Department of Defense was allowed an extra year to reach the 1.25% target and consequently had lower set-aside levels between 1983 and 1986.

Awardees must be independently owned, for-profit firms with less than 500 employees, at least 51% owned by U.S. citizens or permanent residents. Promising proposals are awarded Phase I awards (originally no more than \$50,000; today, \$100,000 or smaller), which are intended to allow firms to determine the feasibility of their ideas. (Typically about 10 Phase I applications are received for every award made.) Approximately one-half of the Phase I awardees are then selected for the more substantial Phase II grants. Phase II awards of at most \$750,000 (originally, \$500,000) are transferred to the small firm as a contract or grant. The government receives no equity in the firm and does not own the intellectual property that the firm develops with these funds. Table 2 displays the annual expenditures on the SBIR program, the number of Phase II awards, and the percentage set-aside.

Conversations with federal program managers, as well as assessments by the U.S. General Accounting Office (1985, 1987*b*, 1995), suggest that the program is to a certain extent insulated from the political pressures to make awards to particular firms encountered in other programs. In particular, three factors limit the pressures to select politically connected firms:

First, the awards are reasonably small in size, not exceeding \$750,000. Consequently, an award is less likely to be the subject of

an intense lobbying activity than that of a program making larger disbursements.

Second, the awards are not made by a single body in each agency but, rather, allocated according to each subunit's external R & D spending. Within the Department of Defense, for instance, hundreds of midlevel officials (whose primary responsibility is overseeing other research and development efforts) decide on SBIR awards. Thus, it is harder to influence the selection process than in programs where a single committee is responsible for making all the awards.

Finally, the scoring systems used to rank applicants largely focus on the technological merit of the proposals, rather than on more subjective assessments of commercial potential. This may leave evaluators with less discretion to choose well-connected applicants.

Despite the presence of these protections, it is not clear that the program always makes awards in a manner that addresses the externalities induced by spillover problems or certifies firms to outside investors. Firms that have participated in the SBIR program, former program managers, and other observers suggest several deviations:

First, the program managers have faced pressure from congressional officials to make geographically diverse awards. The share of funds going to firms in California and Massachusetts—which has consistently been about 40%—has attracted scrutiny from congressmen unhappy about the awards' geographic concentration. One indication of the political sensitivity of the allocation of SBIR awards may be the fact that in almost every recent fiscal year, all 50 states have received at least one SBIR award (U.S. Small Business Administration 1996). In private conversations, several former program managers indicated that interest from congressmen representing districts with few awards went beyond questions about the overall distribution of awards to include in some cases inquiries about the status of particular SBIR applications.

Second, particular companies have demonstrated an ability to capture a disproportionate number of awards. Evaluations of the SBIR program (e.g., U.S. General Accounting Office 1992) suggest that these "SBIR mills"—many of which have staffs in Washington that focus on identifying opportunities for applications—appear to commercialize projects at a significantly lower rate than other firms.

Third, many program officers focus on making awards to firms that are most likely to meet agency needs, with little regard to the effect of awards on the pace of technological innovation or even to the long-run sustainability of the awardees as businesses.

Finally, the contravening pressure exists to back firms that are likely to be successful. As Wallsten (1996) discusses, the Small Business Administration and program officers at many agencies compile stories of successful awardees, which are frequently presented in speeches and

reports. This may lead to pressures to select firms that would succeed even without the award.

IV. The Challenge of Evaluation

While both government agencies and academic economists have sought to assess the efficacy of federal programs to aid technology-intensive industries, they have tended to focus on different questions. Most federal evaluations of these programs have consisted of case studies and surveys of the commercial activity *directly* attributable to the awards. Examples of such evaluations include Myers, Stern, and Rorke (1983), Price, Waterhouse (1985), and U.S. General Accounting Office (1987a, 1989, 1992).

Such surveys of direct commercial effects, however, are subject to several biases. In the case of the SBIR program, small business advocates conducted a protracted lobbying campaign to expand the program. Executives active in this effort may have overstated the benefits from the awards. At the same time, some firms may have downplayed the commercial benefits of the program, lest they attract unwelcome attention from reporters and politicians seeking to expose "corporate welfare." Finally, because many firms pursued SBIR projects closely related to their core technologies, determining the direct effect of an SBIR award may have been difficult.

Studies of federal technology programs by academic economists, beginning with Levy and Terleckyj (1983), have tended to focus on the short-run effects of these efforts. In particular, they often ask whether federal funds substitute for or stimulate private R & D spending. For instance, Irwin and Klenow (1996) show that semiconductor manufacturers substantially reduced their own R & D spending while participating in the Sematech consortium. In the study most closely related to this one, Wallsten (1996) shows that the subset of SBIR awardees that were publicly traded reduced their own spending on R & D in the years immediately following the award.

However valuable a framework when examining the macroeconomic effect of public expenditures, it is less clear that this analytic approach is appropriate when assessing public programs for small high-technology firms. In many cases, small high-technology firms are organized around one key scientist or engineer and his or her research laboratory or product development team. It may not be possible to accelerate the project's progress by "scaling up" the project through the addition of more researchers or technicians. In this case, it is rational for the firm not to increase its rate of spending but rather to use the funds to prolong the time until it needs to seek additional capital. To interpret a short-run reduction in internal research spending by award-

ees as an indication of program ineffectiveness is very problematic. Rather, the long-run effect of the awards is the crucial question.

Once the need for an assessment of the long-term performance of the SBIR awardees was clear, identifying the appropriate metric was still problematic. Ideally, I would have measured both the social and private effects of the program. While case studies of a small number of participants in the SBIR program (along the lines, e.g., of Mansfield et al. 1977) might shed some light on the social benefits, such an effort would have been impossible for a larger sample.⁴

Even determining an appropriate measure of private benefits was difficult. Ideally, I would have examined the relationship between program participation and firms' valuations. Only 2% of the SBIR awardees in the first three program cycles, however, were publicly held at the time of their initial Phase I awards. I focused instead on two alternative measures: sales and employment growth. The valuations assigned to private firms by venture capitalists have been shown to be highly correlated with these two measures (Gompers and Lerner, in press). In a supplemental analysis discussed in Section VIB below, I show that the changes in valuations for the subset of publicly traded firms were consistent with the employment and sales results.

V. The Construction of the Data Set

In this analysis, I relied on a unique database prepared by the GAO. The GAO was mandated in the legislation establishing the SBIR program to evaluate the initiative periodically. In late 1985, the agency generated a sample of 933 firms that received SBIR awards in the first three program cycles from the 12 federal agencies participating in the program at the time. This sample included all firms that had received Phase II awards to that date, as well as a sampling of the firms that had only received the preliminary Phase I awards. The U.S. General Accounting Office surveyed these firms in early 1986 and mid-1988. Eight hundred and thirty-five firms responded to at least one survey. Of these, 541 received Phase II awards in the first three program cycles, while only 294 received Phase I awards.

In addition to the sample of 294 firms that received only Phase I awards, I constructed two matching samples of 300 firms each that did not receive SBIR awards in the first three program cycles. In each case, I selected firms that closely resembled the Phase II awardees using Corporate Technology Information Services' *Corporate Technology Directory* (1996), the most comprehensive directory of U.S. high-

4. These analyses were complemented, however, with eleven interviews of Massachusetts firms that had received at least one Phase II award from the Department of Defense, as well as numerous informal conversations with SBIR program managers.

technology firms. This directory employs a highly detailed industry and geographic classification scheme, which allows quite precise matches.

One set was matched on industry and firm size. I recorded the primary industry classification of each Phase II awardee in the 1986 edition of the *Corporate Technology Directory* or the first subsequent volume in which the firm was listed. (In all, 74% of the Phase II awardees were listed.) I then pulled a random selection of 300 of these awardees. For each selected firm, I identified in the 1986 *Corporate Technology Directory* the firm in the same classification with the closest sales level.⁵ I excluded as possible matches firms that were units or subsidiaries of other concerns or firms that had received any SBIR awards in the first three program cycles (determined through U.S. Small Business Administration [1986]).

The second set was matched on geographic location and firm size. Following the same procedure as above, I identified in the 1986 *Corporate Technology Directory* the firm in the same city and state with the closest sales level to a random sample of 300 of the 541 Phase II awardees. If there were no matches within the same city, I chose a firm from the next city with the same three-digit ZIP code (in the same state) listed in the volume. I determined whether the awardees or matching firms had received venture financing using the Venture Economics and VentureOne databases (described in Gompers and Lerner [1999]).

The samples are summarized in tables 3 and 4. The firms were similar in most respects. The matching firms had slightly greater sales in 1985; the awardees, greater employment. The awardees were slightly more likely to be located in California or Massachusetts (35% vs. 30%).

A final step in constructing the sample was determining the employment and sales of the SBIR Phase II awardees and the matching firms at the end of 1995.⁶ For publicly traded firms, I used Security and Exchange Commission filings. For private firms, I employed a variety of printed and electronic databases.⁷ For firms where I could not ascertain employment and sales, I checked the news stories compiled in various

5. While the standard industrial classification (SIC) scheme places all software firms in class 7372, this directory distinguishes between firms that manufacture educational software used for teaching typing and for music instruction. If there were no matches within the class, I chose a firm in a related industry: e.g., if there was no appropriate match for a firm classified as a plastic composite laminate manufacturer, I matched the firm to another composite laminate manufacturer.

6. In a few cases, firms did not disclose 1985 sales. Furthermore, SBIR awardees were only requested to report ranges for 1985 sales. Consequently, in all cases where firms did not report sales or where reported sales were in one of the ranges above \$1 million, I ascertained 1985 sales using the 1986 editions of the same sources used to determine 1995 sales.

7. Where possible, I used the *Corporate Technology Directory*, one of the few business directories with a rigorous procedure to identify (and correct) problematic responses. I repeated the analyses below, eliminating observations that relied on other commercial data sources. These omissions had little effect on the magnitude of the results.

TABLE 3 Construction of Sample of SBIR Phase II Awardees and Matching Firms

	No. of Firms
General Accounting Office Survey:	
Number of firms surveyed	933
Respondents to 1986 General Accounting Office survey	750
Respondents to 1988 General Accounting Office survey	729
Respondents to either General Accounting Office survey	835
Of respondents to either General Accounting Office survey:	
Received one or more Phase II awards in first three program cycles	541
Did not receive Phase II award in first three program cycles	294
Matching firms selected through 1986 <i>Corporate Technology Directory</i> :	
Industry- and sales-matched	300
Location- and sales-matched	300
Final sample:	
Phase II awardees from survey	541
Matching firms from survey and <i>Corporate Technology Directory</i>	894

TABLE 4 Comparison of SBIR Phase II Awardees and Matching Firms

	Mean	SD	Minimum	Maximum
A. SBIR Phase II awardees:				
Year founded	1977.5	6.9	1935	1985
1985 sales	4.1	8.4	.1	87.9
1985 employment	46.6	85.8	0	600
No. of Phase II awards in first three cycles	2.8	3.5	1	30
Located in California?	.21		0	1
Located in Massachusetts?	.14		0	1
No. of early-stage venture financings, 1983-85, in:				
State	230.0	293.3	0	770
ZIP code	3.2	8.1	0	46
B. Matching firms:				
Year founded	1976.6	9.6	1899	1985
1985 sales	4.2	8.9	.1	86.9
1985 employment	33.2	68.9	0	650
No. of Phase II awards in first three cycles	0	0	0	0
Located in California?	.20		0	1
Located in Massachusetts?	.10		0	1
No. of early-stage venture financings, 1983-85, in:				
State	216.2	286.4	0	770
ZIP code	2.2	6.6	0	46

NOTE.—Panel A provides summary statistics on the 541 firms that received SBIR Phase II awards in the first three program cycles; panel B, the 894 matching firms in the sample. All sales figures are in millions of 1995 dollars.

electronic databases. These searches revealed that many firms for which I was unable to obtain 1995 sales and employment data from the published and on-line directories had gone out of business. In cases where firms were liquidated prior to the end of 1995, I recorded the firm's 1995 employment and sales as zero.

Acquired firms posed another challenge. In many cases, their operations were folded into those of the acquirer, precluding a determination of the original firm's level of activity in 1995. Consequently, I did not include in the analysis the employment and sales of firms that had been acquired unless the companies continued to be operated as separate subsidiaries through the end of 1995. Another exception was cases where the purchase of the firm was an asset sale as part of a bankruptcy, in which case employment and sales were recorded as zero. In some cases, I was only able to determine a range in which sales fell. In these cases, I employed the midpoint of this range.

VI. Empirical Analysis

A. Overall Effect of Subsidies

I first compared the SBIR awardees with the three groups of matching firms. Panel A of table 5 shows that the mean sales increase from the end of 1985 to the end of 1995 was greater for the awardees (\$4.0 million vs. \$1.1 million, both in constant 1995 dollars). Panel B displays a similar employment increase (a boost of 26 employees vs. six). For the mean SBIR awardee, this represented a 98% boost in sales (in inflation-adjusted dollars) and a 56% increase in employment. In both cases, the differences in means were statistically significant.

Both distributions were highly skewed: among the SBIR awardees, for instance, only about one-fifth of the observations were above the mean level of employment growth. This skewness was consistent with evidence on the financial and operating performance of other small firms, such as portfolio companies of venture capitalists or those completing initial public offerings. At each reported percentile, the changes in employment and sales were more positive for the awardees. The differences between the awardees and the matching firms were relatively modest, however, until the top one-third of the distribution. I tested—and rejected—the null hypothesis of the equality of medians, thus demonstrating that a few outliers did not cause the differences.

In unreported analyses, I compared the sales and employment growth between the three subsets of matching firms. These patterns did not appear to be a consequence of peculiarities in one of the groups of matching firms. The growth of sales and employment differed little between the 294 firms that received SBIR Phase I awards and the 600 firms taken from the *Corporate Technology Directory*. For instance,

TABLE 5 Growth of SBIR Phase II Awardees and Matching Firms

	SBIR Phase II Awardees	Matching Firms	<i>p</i> -Value from Comparison
A. Change in sales, 1985-95:			
Mean	4.03	1.14	
90th percentile	8.29	3.68	
75th percentile	1.40	.33	
Median	-.07	-.40	
25th percentile	-.64	-1.00	
10th percentile	-4.01	-7.19	
SE	1.29	.81	
<i>p</i> -value, comparison of means			.046
<i>p</i> -value, comparison of medians			.000
No. of observations	493	836	
B. Change in employment, 1985-95:			
Mean	26.20	5.78	
90th percentile	66	29	
75th percentile	10	2	
Median	-1	-3	
25th percentile	-8	-10	
10th percentile	-45	-50	
SE	10.32	4.47	
<i>p</i> -value, comparison of means			.038
<i>p</i> -value, comparison of medians			.000
No. of observations	499	846	

NOTE.—The table summarizes the change in the sales and employment of the 541 firms that received SBIR Phase II awards in the first three program cycles and the 894 matching firms in the sample. All sales figures are in millions of 1995 dollars.

the mean employment for the Phase I awardees increased by 4.2 jobs, while for the directory firms, the average increase was 6.5. (The average employment increase was 6.2 among the directory firms matched by industry and 6.8 among those matched by geography.) Joint *F*-tests of the null hypothesis that the three groups did not differ, as well as pair-wise comparisons of means and medians, did not identify any significant differences. The differences between the SBIR Phase II awardees that were significant in tables 5 and 6 remained so, at least at the 10% confidence level, when the Phase II awardees were compared only to the Phase I awardees or to the directory firms.

An alternative explanation for the superior long-run growth of the SBIR awardees was that winners cultivated relationships with politicians or federal program managers and were consequently likely to receive subsequent procurement contracts. To examine this possibility, I restricted the above analysis to the two agencies that made significant numbers of SBIR awards but were unlikely to undertake procurement contracts with awardees: the National Science Foundation (NSF) and the Department of Health and Human Services (HHS). (The HHS's external research, and consequently their SBIR awards, overwhelm-

TABLE 6 Growth of SBIR Phase II Awardees and Matching Firms, by Location

Firms Located in a ZIP Code	SBIR Phase II Awardees	Matching Firms	p-Value
A. With an early-stage venture financing, 1983–85:			
Change in sales, 1985–95:			
Mean	7.71	1.06	
Median	-.06	-.42	
SE	3.09	1.23	
p-value, comparison of means			.026
p-value, comparison of medians			.001
No. of observations	189	271	
Change in employment, 1985–95:			
Mean	47.43	3.35	
Median	-1	-4	
SE	22.99	7.99	
p-value, comparison of means			.039
p-value, comparison of medians			.001
No. of observations	190	277	
B. Without an early-stage venture financing, 1983–85:			
Change in sales, 1985–95:			
Mean	1.74	1.18	
Median	-.07	-.40	
SE	.81	1.04	
p-value, comparison of means			.718
p-value, comparison of medians			.000
No. of observations	304	565	
Change in employment, 1985–95:			
Mean	13.14	6.96	
Median	-2	-3	
SE	8.80	5.39	
p-value, comparison of means			.527
p-value, comparison of medians			.057
No. of observations	309	569	

NOTE.—The table summarizes the change in the sales and employment of the 541 firms that received SBIR Phase II awards in the first three program cycles and the 894 matching firms in the sample. Firms are divided by whether their headquarters in 1985 were in a ZIP code with one or more seed or early-stage financings between 1983 and 1985. All sales figures are in millions of 1995 dollars.

ingly came from the basic research-oriented National Institutes of Health.) The gap between the firms that received Phase II awards from these agencies and those that only received Phase I awards was even greater than the dispersion in the sample as a whole. For instance, firms that received Phase II awards from NSF and HHS in the first three program cycles grew by 32.6 employees between 1985 and 1995; those that only received Phase I awards grew by 2.1 employees. These results suggested that the growth of the awardees was not a consequence of them subsequently obtaining procurement contracts.⁸

8. I also compared the awardees from HHS to matching firms specializing in biotechnology and medical devices that had received no SBIR awards, with similar results. An alternative explanation for the superior growth of the SBIR awardees was that winners of SBIR

B. Variation in Subsidy Effect across Regions

I next examined the regional disparities in these growth patterns. To measure new venture activity in the region where the firm was located, I tabulated seed and early-stage financings by private venture capital funds in the years 1983–85. I determined the dollar volume and number of early-stage financings in each five-digit ZIP code using the Venture Economics database.⁹

Panels A and B of table 6 present comparisons similar to those in table 5 but also divide firms by whether their headquarters at the end of 1985 was in a ZIP code with at least one early-stage venture financing in the years 1983–85. The differences between the SBIR awardees and the matching firms in the ZIP codes with venture capital activity were pronounced. For instance, employment increased by 47 for these awardees, as opposed to increasing by three for the others. For the average SBIR awardee in a ZIP code with venture activity—which in 1985 had \$5.1 million in sales (again in 1995 dollars) and 57 employees—inflation-adjusted sales over the next decade grew by 151% and employment by 83%. The differences were much smaller in other areas (employment rose by 13 for the awardees, as opposed to by seven for the matching firms). The differences in means were statistically significant only among the firms in regions with venture activity. Medians displayed a similar pattern. The difference in median sales, however, was also highly significant when awardees and nonawardees were compared in areas without venture capital activity.

Table 7 examines the growth of employment and sales in a regression framework. The table examines the change in employment and sales, as well as the percentage change in these variables.¹⁰ Among the independent variables were 1985 sales or employment, the volume (in millions of dollars) of early-stage venture capital activity in the firm's ZIP code between 1983 and 1985, a dummy variable indicating whether the firm received any Phase II awards in the first three SBIR funding

awards were more likely to win subsequent SBIR awards. While this may have been true in the initial years of the program (e.g., U.S. General Accounting Office 1987b), conversations with federal officials suggest that it has been much less of an issue in recent years. Not only were SBIR awards to firms that had received multiple previous grants more likely to be scrutinized by the GAO, resulting in a reluctance to make such grants, but the 1992 reauthorization of the SBIR program (Public Law 102-564) added additional criteria for the evaluation of SBIR applications of firms that had received awards in the past.

9. Five-digit ZIP codes are a fine gradation: the city of Cambridge, Massachusetts, for instance, is divided into six five-digit ZIP codes. I repeat the analyses in Section VI using three-digit ZIP codes (the code 021 encompasses the city of Boston and the nearby suburbs). I also repeated the analyses using the *number* of seed and early-stage financings in the ZIP code. These alternative measures had little effect on the results. In an analysis later in this section, I explore the use of broader geographic divisions.

10. Because some of the firms had minimal employment and sales in 1985, I calculated the percentage change variable as $100 * (1995 \text{ value} - 1985 \text{ value}) / (1985 \text{ value} + 1)$.

TABLE 7 **Regression Analyses of the Change in Employment and Sales between 1985 and 1995**

A. OLS Regressions, with Change and Percentage Change in Sales as the Dependent Variables

	Change	% Change
1985 sales	-.18 [2.67]	-2.28 [2.46]
Value of early-stage financings in ZIP code	.07 [2.12]	1.90 [2.56]
Any Phase II SBIR awards in first three cycles?	1.88 [.80]	25.05 [.80]
Value of financings * any Phase II awards in first three cycles	.16 [4.75]	2.41 [3.43]
Constant	1.63 [1.26]	60.34 [2.07]
Root mean square error	25.37	564.93
Adjusted R^2	.02	.01
No. of observations	1,329	1,329

B. OLS Regressions, with Change and Percentage Change in Employment as the Dependent Variables

	Change	% Change
1985 employment	-.19 [3.32]	-.46 [5.13]
Value of early-stage financings in ZIP code	.29 [.85]	.39 [.31]
Any Phase II SBIR awards in first three cycles?	14.70 [1.00]	-12.54 [.41]
Value of financings * any Phase II awards in first three cycles	1.43 [11.84]	5.11 [5.45]
Constant	10.83 [3.80]	110.08 [4.89]
Root mean square error	173.04	930.47
Adjusted R^2	.02	.002
No. of observations	1,345	1,345

C. Median Regressions, with Change and Percentage Change in Sales as the Dependent Variables

	Change	% Change
1985 sales	-.62 [5.37]	-1.69 [5.02]
Value of early-stage financings in ZIP code	.003 [4.09]	.11 [.43]
Any Phase II SBIR awards in first three cycles?	.39 [18.42]	18.96 [3.08]
Value of financings * any Phase II awards in first three cycles	.01 [10.67]	.27 [.69]
Constant	-.03 [1.89]	-17.38 [4.37]
Sum of absolute weighted deviations	7,918.87	160,923.20
Pseudo R^2	.07	.01
No. of observations	1,329	1,329

D. OLS Regressions, with Change and Percentage Change in Sales as the Dependent Variables, Using Only Firms Located in ZIP Codes without Early-Stage Venture Financings

	Change	% Change
1985 sales	-.20 [3.91]	-1.75 [4.23]
Was state's venture capital financings above mean?	-1.07 [.65]	-36.75 [.94]
Any Phase II SBIR awards in first three cycles?	-.53 [.37]	-8.50 [.21]
State venture dummy * any Phase II awards in first three cycles	2.56 [1.98]	84.57 [6.10]
Constant	2.38 [1.26]	69.61 [1.89]
Root mean square error	21.62	571.10
Adjusted R^2	.01	.002
No. of observations	869	869

TABLE 7 (Continued)

E. OLS Regressions for Firms in Industries with High and Low Intangible to Total Asset Ratios, with Change in Sales as the Dependent Variable

	Industry Intangible to Total Asset Ratio	
	Above Median	Below Median
1985 sales	-.13 [1.64]	-.29 [4.89]
Value of early-stage financings in ZIP code	.12 [1.62]	.02 [1.67]
Any Phase II SBIR awards in first three cycles?	3.26 [.96]	-.07 [.04]
Value of financings * any Phase II awards in first three cycles	.44 [5.73]	-.08 [1.01]
Constant	.90 [.98]	3.19 [1.61]
Root mean square error	24.97	25.67
Adjusted R^2	.05	.002
No. of observations	789	531
p -value, F -test of equality of SBIR coefficients in two regressions		.000

F. OLS Regressions for Firms in Industries with High and Low R & D to Sales Ratios, with Change in Employment as the Dependent Variable

	Industry R & D/Sales Ratio	
	Above Median	Below Median
1985 employment	-.15 [1.63]	-.19 [7.51]
Value of early-stage financings in ZIP code	.10 [.21]	.60 [1.67]
Any Phase II SBIR awards in first three cycles?	10.13 [.60]	19.50 [1.14]
Value of financings * any Phase II awards in first three cycles	3.60 [6.71]	-1.19 [2.31]
Constant	16.08 [1.69]	3.98 [.60]
Root mean square error	207.27	116.07
Adjusted R^2	.03	.02
No. of observations	735	600
p -value, F -test of equality of SBIR coefficients in two regressions		.000

G. OLS Regressions with Change and Percentage Change in Sales as the Dependent Variables, and Additional Independent Variables for Multiple SBIR Awardees

	Change	% Change
1985 sales	-.19 [2.64]	-2.28 [2.28]
Value of early-stage financings in ZIP code	.07 [2.13]	1.90 [2.56]
Any Phase II SBIR awards in first three cycles?	.11 [.08]	19.83 [1.60]
Multiple Phase II SBIR awards in first three cycles?	4.02 [1.66]	15.61 [.99]
Value of financings * any Phase II awards in first three cycles	.48 [2.69]	8.71 [2.22]
Value of financings * multiple Phase II awards in first three cycles	-.58 [2.24]	-10.93 [1.93]
Constant	1.67 [1.27]	60.35 [2.05]
Root mean square error	25.23	562.73
Adjusted R^2	.02	.01
No. of observations	1,329	1,329
p -value, F -test of insignificance of multiple awardee variables	.018	.144

TABLE 7 (Continued)

H. OLS Regressions, with Change and Percentage Change in Employment as the Dependent Variables, and Additional Independent Variables for Multiple SBIR Awardees

	Change	% Change
1985 employment	-.19 [3.05]	-.42 [5.86]
Value of early-stage financings in ZIP code	.29 [.85]	.38 [1.30]
Any Phase II SBIR awards in first three cycles?	11.72 [.79]	.59 [.02]
Multiple Phase II SBIR awards in first three cycles?	8.05 [.78]	-23.99 [2.44]
Value of financings * any Phase II awards in first three cycles	3.96 [7.41]	13.46 [4.40]
Value of financings * multiple Phase II awards in first three cycles	-4.39 [4.19]	-14.25 [3.16]
Constant	10.75 [4.08]	108.89 [4.74]
Root mean square error	171.81	928.16
Adjusted R^2	.03	.01
No. of observations	1,345	1,345
<i>p</i> -value, <i>F</i> -test of insignificance of multiple awardee variables	.000	.007

NOTE.—The dependent variables in the regressions are the change and percentage change in sales (in millions of 1995 dollars) and employment between the end of 1985 and the end of 1995. The percentage change is computed as $100 * [(1995 \text{ value} - 1985 \text{ value}) / (1985 \text{ value} + 1)]$. The independent variables are sales (in millions of 1995 dollars) and employment at the end of 1985, the volume (in millions of dollars) of seed and early-stage venture capital transactions between 1983 and 1985 in the ZIP code in which the firm was located in 1985, a dummy variable measuring whether the firm received one or more Phase II SBIR awards in the first three program cycles, and an interaction between the venture capital and SBIR variables. Panel A presents ordinary least squares (OLS) regressions with change and percentage change in sales as the dependent variables; panel B presents similar OLS employment regressions. Panel C uses a median regression specification to analyze the change and percentage change in sales. Panel D uses only those firms based in ZIP codes without a seed and early-stage venture capital transaction between 1983 and 1985. This regression uses a dummy variable denoting whether the state had above the mean level of *per capita* venture financings instead of the dollar volume of disbursements in the ZIP code. Panels E and F analyze the change in sales and employment, estimating separate OLS regressions for firms whose industries are above or below the median in intangible to total assets and R & D to sales ratios at the end of 1985. *F*-tests examine the null hypothesis that the coefficients on the SBIR variables are the same across the sub-samples. Panels G and H add to the basic OLS specification a dummy variable denoting whether the firm had received more than one SBIR award in the first three program cycles, and an interaction between this and the venture capital activity in the firm's ZIP code between 1983 and 1985. *F*-tests examine the null hypothesis that the two new variables do not differ from zero. The sample (in all regressions except panel D) consists of 541 firms that received SBIR Phase II awards in the first three program cycles and 894 matching firms. Absolute *t*-statistics (computed using heteroskedastic-consistent standard errors in all but panel C) are in brackets.

cycles, and an interaction between the venture capital activity and SBIR awardee variables.

In panel A, I examined the change in sales and the percentage change in sales (both in constant 1995 dollars) in ordinary least squares (OLS) regressions. Because the firms were of very different sizes, a heteroskedasticity problem potentially existed. Consequently, I divided the firms into groups on the basis of 1985 sales and calculated heteroskedastic-consistent standard errors. (A similar procedure was used in all the OLS regressions reported in this table.) Panel B undertakes a similar analysis of the change in employment.

The regressions strongly supported the suggestion that the superior

performance of SBIR awardees was not uniform. The presence of an SBIR award alone had little relationship with sales and employment growth. Rather, only the interactions between the SBIR indicator and venture activity in the ZIP code were consistently significant. The SBIR awards were associated with firm growth in areas with considerable venture financing activity, but no relationship emerged elsewhere. These patterns were consistent with suggestions that politically connected firms with few growth prospects may have captured awards in regions without large numbers of high-technology firms or venture capital organizations.

While the coefficient of the interaction variable was consistently significant, the overall goodness-of-fit in the regressions was disappointing. The poor fit reflected the fact that much of the success and failure of small firms is driven by idiosyncratic, firm-specific factors, difficult to control for in a large-sample study. This pattern raised the concern, however, that a few extreme outliers might drive the results. In order to address this issue, I employed a median regression specification, which reduced the influence of outliers. As panel C of table 7 reports, these modifications did have some effect on the results: the magnitude and statistical significance of the coefficient of the SBIR awardee dummy variable increased, while the interaction term became less important. This pattern was consistent with the median analysis of sales reported in table 6, which suggested a significant difference between awardees and nonawardees across all regions.

If the geographic disparities were a consequence of political pressures, then we might expect also to see differences among the firms located in ZIP codes without venture capital. Political pressures to allocate SBIR funds to firms located in states with high levels of venture capital activity (even if concentrated elsewhere) should have been considerably less than in states with very little venture activity at all. Put another way, state and congressional officials were much more likely to conduct a concerted lobbying campaign for awards to firms in Idaho than in northeastern California.

To address this suggestion, I undertook regression analyses confined to the SBIR awardees and matching firms that were located in ZIP codes without a seed or early-stage venture capital financing round between 1983 and 1985. I compared the awardees and matching firms located in those states with above and below the mean level of per capita venture capital disbursements during this 3-year period. Panel D of table 7 presents two regression analyses, which employ the change in sales as a dependent variable, as well as the percentage change. The independent variables were 1985 sales of the firm, a dummy variable indicating whether the firm received any Phase II awards in the first three SBIR funding cycles, a dummy variable denoting whether the state received above the mean amount of per capita venture capital

disbursements in the years 1983–85, and an interaction between the dummy variables denoting SBIR awardees and above-average venture capital activity in the state.

The political pressure hypothesis suggests that the disparity between awardees and matching firms should have been greater in the states with high levels of venture capital activity. In states with low levels of venture activity, the distortions in the award process may have led to the selection of firms that could not benefit from the funds. Consistent with this suggestion, the interaction between the SBIR awardee dummy and that denoting above-average venture capital activity was significantly positive. Consider, for instance, the coefficient of 2.56 on the interaction term in the leftmost regression. This implied that the difference in the sales growth of the SBIR awardees and the matching firms (in ZIP codes without venture capital activity) was \$2.6 million greater in states with above the mean level of venture activity.

I examined the robustness of the geographic regressions in several additional unreported analyses. First, I was concerned that the results could be an artifact of the two measures of firm success that I used, employment and sales. I consequently examined the changes in valuations, under a quite stark assumption: that the valuation of privately held or acquired firms was nominal when compared to the publicly traded firms. If this supposition held, changes in public market valuations could provide a reasonable indication of the overall change in these firms' valuations. I summed the value of those SBIR awardees and matching firms that were publicly traded in 1985 and 1995 (determining these valuations through the databases of the Center for Research in Security Prices and the several *Moody's* stock guides). I then examined the difference between the total valuation at the beginning and end of this period. The difference in the growth of the SBIR awardees and the matching firms was dramatic. The increase in the public market valuation of the 541 SBIR awardees between 1985 and 1995 was \$9.5 billion. The valuation of the 894 matching firms, in contrast, only increased by a total of \$3.0 billion during this period. Consistent with the results involving employment and sales, the increase in the valuation was the greatest for the awardees that were located in a ZIP code with an early-stage venture financing in the years 1983 to 1985. The 209 awardees located in such ZIP codes increased in value by \$8.7 billion; the 332 awardees located in other ZIP codes increased in value by under \$800 million.

Second, I was concerned about my matching approach. As discussed above, I constructed two populations of matching firms. An alternative approach would have been to undertake a single matched set, based on both region and industry: that is, rather than matching a Cambridge genome sequencing company to either another Cambridge firm or another genome firm, I could have selected a matching firm in the same

industry and region. I chose not to employ this procedure owing to fears that it would introduce systematic biases. In particular, when I attempted to do such a matching on two criteria, I found that the quality of the matches differed dramatically by region. While quite precise matches could be obtained in California and Massachusetts, in some other regions the matching firms had very different characteristics (e.g., they tended to be much larger than the awardees). This was a consequence of the fact that there were few potential matches in many states: for example, in Hawaii, Idaho, and Kentucky, there were only a handful of entries in the 1986 *Corporate Technology Directory*. I was concerned that such a procedure would bias the results in ways that would be very difficult to address in the subsequent analyses. I undertook various tests in order to address concerns that one of the subpopulations of matching firms that I constructed might still be biased in some not readily apparent manner. I thus reestimated the equations only using the Phase II SBIR awardees and the *Corporate Technology Directory* matching firms, as well as only the Phase II and Phase I SBIR awardees. I similarly used only the *Corporate Technology Directory* firms matched on industry and those matched on location. The smaller sample sizes led to lower significance levels but had little effect on the magnitudes or signs of the coefficients. I also added controls for the age of the firm, whether it was venture-backed by the end of 1985, and for the two-digit SIC code of the firm, in case the results were shaped by some unobserved differences between the SBIR awardees and the matching firms. These had little effect on the coefficients or significance levels.

Third, I explored the use of alternative econometric specifications. Because the dependent variable in the regressions analyzing the percentage change in sales and employment was bounded at -100% (corresponding to the liquidation of a firm), I repeated these analyses employing a Tobit specification. This had little effect. The magnitude and significance of the coefficients changed little when I estimated systems of two equations, with the second equation examining the propensity of firms to win Phase II awards in the first three program cycles. I also estimated a two-stage Heckman regression, looking first at whether the firm survived and then at the extent of its growth. The interaction terms remained strongly significant. An alternative way to control for outliers was to divide the dependent variable in each regression into 10 deciles: that is, I created 10 equal-sized groupings based on the change in sales and employment. The results with this ordered logit specification very closely resembled those in the median regressions reported above.

C. Industry Patterns

The next analysis examined whether these patterns were especially pronounced in certain industries. As discussed above, the contrasting ex-

planations for the growth of the SBIR awardees—the suggestions that SBIR awards certified firms to investors and that federal officials attempted to select successful firms for SBIR awards, whether the funds were needed or not—implied different empirical patterns. To address these concerns, I divided the firms into those in higher- and lower-technology industries and reestimated the regressions in panels A and B of table 7.

To divide the firms, I examined the characteristics of the firm's industry. Because I could not observe the financial condition of the private firms, I used the characteristics of the average public firm in the same industry. I computed the ratios of intangible assets to total assets, R & D spending to sales, and market to book equity values for all firms with a primary assignment in Compustat to the same four-digit SIC class as a company in the sample during the fiscal year ending in 1985 or at the end of the fiscal year ending in 1985. If there were fewer than four firms assigned to a four-digit SIC code, I used all firms with a primary assignment to the same three-digit SIC code. I then computed the average of these ratios.

I present the results of two of the analyses in panels E and F of table 7.¹¹ I report separate regressions for the observations from industries above and below the median on these proxies. I also present the *p*-values from *F*-tests of the null hypothesis that the two coefficients measuring the effect of SBIR awards (the dummy variable and the interaction term) were identical in the two regression analyses. In the regression employing the change in sales as the dependent variable (panel E), the SBIR dummy variable was positive, and the interaction term was positive and significant, for the firms in industries with more intangible assets. Among the other firms, the two variables were negative and insignificant. The null hypothesis of no difference was rejected. In panel F, where the change in employment was used as the dependent variable and firms were divided by industry R & D to sales ratios, a similar pattern held for firms with R & D spending above the median. The null hypothesis of no difference was once again rejected. This pattern was consistent with the view that SBIR awards provided certification to firms.

D. Number of Awards

The fourth empirical analysis examined how the effect of the awards changed with the number of awards. The distribution of awards in the first three program cycles was quite skewed. While the median SBIR Phase II awardee firm received one Phase II award in the first three

11. The results dividing firms on the basis of the third measure (the ratio of market to book equity values) was similar. The analyses were also robust to checks described in the final paragraphs of Section VIB.

program cycles, four firms received a total of 90 awards. (The mean firm received 2.8 awards in this period.) I added two independent variables that controlled for firms that received multiple SBIR awards. As discussed above, if the main role of public subsidies is to address the externality created by small firms' inability to capture the bulk of the rents from their innovations, more awards should translate into faster growth. As long as the firms capture some of surplus generated by the research, they should continue to benefit from additional subsidies. The certification hypothesis suggests a more ambiguous relationship, with the marginal value of subsequent awards declining sharply. Furthermore, the pursuit of these awards might introduce increasing distortions as the size (or number) of subsidies increases.

Panels G and H of table 7 present four regressions. The regressions were identical to those in panels A and B, with the addition of a dummy variable denoting firms that received more than one Phase II award in the first three program cycles and an interaction between this new dummy and the measure of venture activity in the firm's ZIP code. I also present the p -values from F -tests of the null hypothesis that these two new variables were jointly equal to zero. The dummy denoting frequent awardees was never significantly positive and in one regression was negative in sign. The interaction term between the multiple awardee dummy and the amount of venture activity in the ZIP code was significantly negative in all regressions. The joint test of the two variables rejected the null hypothesis of no difference from zero in three out of the four regressions.¹² The pronounced superior performance of the SBIR awardees in regions with venture activity was confined to firms receiving modest subsidies. Additional awards appear to have had minimal positive benefits, and the pursuit of these awards may even have had detrimental effects on the firms.

E. SBIR Awards and Venture Capital Financing

An important group to whom SBIR awards may provide information is the venture capital community. As discussed above, venture capitalists fund only a handful of the high-technology businesses begun each year. While the number of companies funded in this manner is small, the amount of capital provided is significant. Furthermore, these firms account for a disproportionately large share of the companies going public each year (Gompers and Lerner 1999) and of successful patent applications (Kortum and Lerner 1998). The importance of venture capitalists suggests an alternative test of the certification hypothesis. In this section, I examine not the overall growth of the SBIR awardees

12. The results were robust to using other cut-off points to denote multiple SBIR awardees, such as firms that received more than five Phase II awards. In unreported analyses, I also examined the robustness of the results to the controls described in Section VIB.

and the matching firms but, rather, the relationship between the receipt of SBIR awards and venture capital financing.

The interpretation of the results may be quite different, depending on the relative timing of the awards and the venture financing. If the SBIR awards played a certification function, then we might expect to see awardees disproportionately receiving venture capital financing in the years after the awards. The awards themselves might have served as a signal to venture capitalists or, alternatively, might have allowed the firm to attract high-quality managers or individual investors who in turn attracted venture investors. It would be harder to argue that the awards played a certification role if awardees were no more likely to receive venture financing than matching firms. The reverse relationship would also be problematic: that is, if venture capital financing *predicted* the receipt of an SBIR award. This might suggest—in the spirit of the models of political capture delineated in Section II above—that the venture capitalists were using the SBIR program as a subsidy for firms already in their portfolios. Rather than refinancing firms, they may have pushed them to obtain public funds.

Table 8 examines these issues in univariate and regression analyses. Panel A presents the probability that the awardees and the matching firms received venture financing in the years before and after the first three program cycles. First, I present the probability that the firms received venture financing through the end of 1981, the year immediately previous to the first SBIR awards. While the probability that SBIR awardees received venture financing was greater, the difference was not statistically significant. I then present the probability that the firms received venture financing for the first time after 1987, when the results of the first three award cycles were largely known. (Awardees were typically notified of whether they received a Phase II award within a year of completing their Phase I project.) In this calculation, I excluded firms that had already attracted venture financing or were publicly traded by the end of 1987. The difference between the awardees and the matching firms was larger in magnitude and statistically significant. These patterns were robust to the selection of other break points (e.g., 1980, 1982, and 1986). Thus, the univariate comparisons suggested that the awards might have certified firms to venture capitalists.

One concern with these univariate results is that they may reflect unobserved differences between the firms. In order to address these concerns, I examined these patterns in logit regression analyses. I first examined whether the presence of a venture capital investor explained the probability that a firm received a SBIR award. I used each of the 1,435 firms as an observation. The dependent variable was a dummy that took on the value of one if the firm received an SBIR Phase II award in the first three program cycles. Independent variables were a dummy denoting whether the firm received venture financing through

TABLE 8 **Univariate and Regression Analyses of SBIR Awards and Venture Financings****A. Univariate Analysis of Probability of Venture Capital Financings for First Time**

	SBIR Phase II Awardees (%)	Matching Firms (%)	<i>p</i> -Value
Funding prior to end of 1981	2.2	1.3	.210
Funding after 1987	2.5	.9	.020

B. Logit Regression Analyses

	Dependent Variable	
	Did Firm Receive SBIR Award in First Three Cycles?	Did Firm Receive Venture Funding after 1987?
Value of early-stage financings in ZIP code	.01 [2.04]	-.02 [.77]
Was firm based in California?	-.001 [.00]	.11 [.16]
Was firm based in Massachusetts?	.25 [1.27]	.62 [.88]
Did firm receive venture capital prior to end of 1981?	.34 [.73]	
Any Phase II SBIR awards in first three cycles?		1.39 [2.60]
1985 employment		.002 [.54]
χ^2 -statistic	173.33	24.01
<i>p</i> -value	.000	.020
Log likelihood	-431.46	-81.35
No. of observations	1,435	1,193

NOTE.—Panel A summarizes the probability that the SBIR awardees and the matching firms received venture capital financing. The panel presents the probability that the firms received venture financing for the first time before the end of 1981 and after 1987. (Firms that received venture capital financing or went public in 1987 or before are excluded from the second analysis.) Panel B presents two logit regression analyses. The dependent variable in the first regression is a dummy coded as 1.0 if the firm received an SBIR Phase II award in the first three program cycles. The dependent variable in the second regression is a dummy coded as 1.0 if the firm received venture financing for the first time after 1987. The independent variables are the volume (in millions of dollars) of seed and early-stage venture capital transactions between 1983 and 1985 in the ZIP code in which the firm was located in 1985, dummy variables denoting companies based in California and Massachusetts in 1985, dummy variables denoting the firm's two-digit standard industrial classification code (not reported), a dummy variable denoting whether the firm received venture financing prior to the end of 1981 (in the first regression only), a dummy variable measuring whether the firm received one or more SBIR Phase II awards in the first three program cycles (in the second regression only), and the firm's employment at the end of 1985 (in the second regression only). The sample consists of 541 firms that received SBIR Phase II awards in the first three program cycles and 894 matching firms. (Firms that received venture capital financing or went public in 1987 or before are excluded from the second regression.) The first regression employs a fixed-effect logit specification; the second, a standard logit specification. Absolute *t*-statistics are in brackets.

the end of 1981 and controls for whether the firm was located in Massachusetts or California, the extent of venture capital activity in the ZIP code, and the two-digit SIC code. Because I constructed the data set by matching the universe of SBIR awardees to a sample of nonawardees, a simple logit analysis would have led to inconsistent and biased estimates (Palepu 1986). To address this problem, I estimated a fixed-effects logit regression (Chamberlain 1980). As the leftmost regression in panel B reports, once the firm and industry characteristics were controlled for, the presence of an earlier venture capital round did not predict whether the company received an SBIR award. The results were robust to modifications in the set of regressors.

I then examined the probability that the firm received venture capital financing for the first time in the years after the three SBIR award cycles (i.e., after 1987). I excluded from the analysis any firms that received venture financing or went public in earlier years.¹³ I once again controlled for the characteristics of the firm (including now the employment level at the end of 1985) and added a dummy variable denoting whether the firm had received any Phase II SBIR awards in the first three program cycles. In the reported regression—as well as in others using modifications of the dependent and independent variables—the SBIR awardee variable was significantly positive. The SBIR awardees were more likely to receive venture financing. At the mean of the other independent variables, the predicted probability of a nonawardee obtaining venture financing was 0.8%, while that of an SBIR awardee was 3.1%.

VII. Conclusions

This article examined the long-run performance of high-technology firms receiving funds from a major public venture capital initiative, the SBIR program. I compared the growth of awardees to a set of matching firms. The SBIR awardees enjoyed substantially greater employment and sales growth. While the awardees and matching firms did not differ significantly in the likelihood of receiving venture capital in the years prior to the awards, in subsequent years the awardees were significantly more likely to receive such financing.

This pattern, however, was not uniform. The superior growth of SBIR awardees was confined to firms based in ZIP codes with substantial venture capital activity. These patterns were more pronounced in high-technology industries. No increase of performance was associated with larger subsidies. These patterns were consistent with the awards playing an important role in certifying firm quality but also with some

13. The sample also shrank because of missing employment data and because some independent variables completely determined the outcome.

distortions of the award process. These findings raised questions about the practice of awarding large numbers of SBIR awards to a few firms, as well as about congressional efforts to insure greater geographic dispersion of awards.

Two important limitations should be acknowledged, which in turn point to future research opportunities.¹⁴ First, this analysis has not sought to assess the social benefits of the program.¹⁵ Numerous studies have suggested that, because of knowledge spillovers, social rates of return to R & D are often much higher than the private returns that the firms performing the research enjoy. This analysis has focused exclusively on private returns, as roughly measured through sales and employment growth. The differentials between the private and social benefits of the SBIR awards might be particularly large because many projects involve very early-stage technologies (where spillovers to other firms may be more frequent) or those important to national defense or public health. The detailed case studies prepared by the National Science Foundation (Tibbetts 1996) suggest that awards in many cases led to substantial spillovers. At the same time, the gains in employment and sales of the typical awardee could have come at the expense of small losses for many competitors. Furthermore, many academic critics (e.g., Mervis 1996) argue that the SBIR set-aside has led to a reduction in funding for academic research, which may have even greater social benefits. While challenging, a detailed computation of the social costs and benefits must be a crucial element in assessing these programs.

14. Another issue is that political pressures and economic changes may have led to a deterioration of the SBIR program's effectiveness over time. For instance, as discussed above, the set-aside for SBIR awards has increased dramatically. This may have led to more efforts to influence the award process. The pool of venture capital also greatly increased during the 1980s and 1990s. The expansion of the private equity market may have obviated the need for such a program, and the long-run competitive advantage that SBIR awards confer today may be much lower. No performance differential can be seen in the relative success of firms that received SBIR awards in the first and third program cycles (venture disbursements had increased sharply between 1982 and 1984). But questions about the changes in the effectiveness of the program over longer time spans will only be answerable in the future.

15. While a rough calculation indicates that this program has had an attractive return for the federal government, these estimates must be approached with a degree of skepticism. For instance, given that the typical SBIR awardee 10 years later has \$2.9 million in additional sales and 20 more employees than a matching firm, under plausible assumptions (30% corporate and individual tax rates, \$50,000 annual salary per employee, and 15% pretax profit margins), the federal government would receive in additional taxes in 1 year an amount almost equal to the (nominal) amount of a Phase II award. If this level of enhanced sales and employment were sustained for a little more than 2 years, the government's rate of return would exceed the 10-year Treasury bond rate. Any more sustained increase in the firm's activity would make the government's return even more attractive. The returns would increase if there were significant spillovers to other firms and consumers or decrease if much of the growth was at the expense of other firms (resulting in a smaller marginal effect on aggregate economic activity or employment).

A second important unanswered question relates to the design of public venture capital programs such as the SBIR initiative. Because this study has focused on a single program that has been essentially unchanged since its inception, it is difficult to assess whether the program could be altered in significant ways. For instance, in light of evidence suggesting that a key benefit of SBIR awards is the certification that they provide, it might be possible that a program that offered much more modest subsidies could also be effective in certifying the quality—and spurring the growth—of small high-technology firms. Two crucial assumptions, however, are that such recognition would be sufficient to motivate firms to seek awards and that the integrity of the selection process could be insured. The recent wave of experimentation with public venture capital programs, both in the United States and abroad, suggests that a cross-program analysis of these and other questions could yield important insights.

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